

C05- ASSESSMENT TOOL -2

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COURSE :Cryptography and network security

COURSE CODE :CSA5113

1. Write pseudocode to simulate a simplified TLS handshake process, including client request, server authentication using a digital certificate, session key generation, and secure data exchange.

ANSWER:

// Step 1: Client sends connection request

CLIENT:

Generate random value ClientRandom

Send "ClientHello" with:

TLS version

Supported cipher suites

ClientRandom

// Step 2: Server responds

SERVER:

Generate random value ServerRandom

Select TLS version and cipher suite

Send "ServerHello" with:

Selected TLS version

Selected cipher suite

ServerRandom

// Step 3: Server authentication

SERVER:

Send Digital Certificate to CLIENT

CLIENT:

Verify the server certificate

Check certificate validity

Check certificate issuer

Check certificate expiration

IF certificate is valid THEN

Continue handshake

ELSE

Terminate connection

ENDIF

// Step 4: Generate session key

CLIENT and SERVER:

Perform secure key exchange

Generate shared secret

Derive session key from:

ClientRandom

ServerRandom

Shared secret

// Step 5: Confirm secure connection

CLIENT:

Send "Finished" message encrypted with session key

SERVER:

Verify the Finished message

Send "Finished" message encrypted with session key

// Step 6: Secure data exchange

CLIENT:

Encrypt application data using session key

Send encrypted data to SERVER

SERVER:

Decrypt received data using session key

Process the data

SERVER:

Encrypt response using session key

Send encrypted response to CLIENT

CLIENT:

Decrypt response using session key

// Secure communication established

Display "TLS Handshake Successful"

Display "Secure Data Exchange Completed"

2. Develop pseudocode for a firewall rule engine that filters incoming packets based on IP address, port number, and protocol, and decides whether to allow or block traffic.

ANSWER:

// Define firewall rules

RULES:

Rule 1: Allow source IP = 192.168.1.10,

Port = 80,

Protocol = TCP

Rule 2: Allow source IP = 192.168.1.20,

Port = 443,

Protocol = TCP

Rule 3: Block source IP = 10.0.0.5,

Port = ANY,

Protocol = ANY

Rule 4: Block all other traffic

// Receive incoming packet

RECEIVE PACKET

Extract:

SourceIP

DestinationIP

Port

Protocol

// Check packet against firewall rules

FOR each RULE in RULES

IF SourceIP matches RULE.SourceIP

AND Port matches RULE.Port

AND Protocol matches RULE.Protocol THEN

IF RULE.Action = "ALLOW" THEN

Display "Packet Allowed"

FORWARD packet

STOP

ELSE IF RULE.Action = "BLOCK" THEN

Display "Packet Blocked"

DROP packet

STOP

ENDIF

ENDIF

END FOR

// Default action

Display "Packet Blocked by Default Rule"

DROP packet

3. Write pseudocode to detect and classify malware behavior (virus, worm, or Trojan) based on propagation method, file modification patterns, and user interaction.

ANSWER:

// Receive malware behavior information

INPUT MalwareBehavior

Extract:

PropagationMethod

FileModification

UserInteraction

// Initialize classification

MalwareType = "Unknown"

// Check for Virus

IF PropagationMethod = "Requires user action"

AND FileModification = "Modifies or infects files" THEN

MalwareType = "VIRUS"

// Check for Worm

ELSE IF PropagationMethod = "Spreads automatically through network"

AND UserInteraction = "Not required" THEN

MalwareType = "WORM"

// Check for Trojan

ELSE IF PropagationMethod = "Does not self-replicate"

AND UserInteraction = "Requires user to run or install program"
THEN

MalwareType = "TROJAN"

ELSE

MalwareType = "UNKNOWN MALWARE"

ENDIF


```
// Display classification
```

```
Display "Malware Classification: ", MalwareType
```

4. Design pseudocode for an Intrusion Detection System (IDS) that uses both signaturebased and anomaly-based detection to identify suspicious network activity.

ANSWER:

```
// Load known attack signatures
```

```
Load SignatureDatabase
```

```
// Collect normal network behavior
```

```
Establish NormalTrafficProfile
```

```
WHILE Network is Active
```

```
    // Capture incoming network packet
```

```
    Capture Packet
```

```
    Extract:
```

```
        SourceIP
```

```
        DestinationIP
```

```
        SourcePort
```

```
        DestinationPort
```

```
        Protocol
```

PacketSize

TrafficRate

PacketContent

SignatureMatch = FALSE

AnomalyDetected = FALSE

// Signature-Based Detection

FOR each Signature in SignatureDatabase

IF Packet matches Signature THEN

SignatureMatch = TRUE

Display "Known Attack Detected"

Generate Alert

BREAK

ENDIF

END FOR

// Anomaly-Based Detection

IF TrafficRate is significantly different from NormalTrafficProfile

OR PacketSize is abnormal

OR SourceIP shows unusual behavior

OR DestinationPort is unusual

OR Protocol usage is abnormal THEN

AnomalyDetected = TRUE

Display "Anomalous Activity Detected"

Generate Alert

ENDIF

// Final Classification

IF SignatureMatch = TRUE AND AnomalyDetected = TRUE
THEN

Display "High Risk: Known Attack with Abnormal Behavior"

Take Security Action

ELSE IF SignatureMatch = TRUE THEN

Display "Threat Type: Known Attack"

Take Security Action

ELSE IF AnomalyDetected = TRUE THEN

Display "Threat Type: Suspicious/Unknown Activity"

Take Security Action

ELSE

Display "Network Activity is Normal"

ENDIF

END WHILE

5. Create pseudocode for an Intrusion Prevention System (IPS) that monitors network traffic in real time and automatically blocks malicious requests while logging intrusion attempts.

ANSWER:

// Load security rules and known attack signatures

Load SignatureDatabase

Load SecurityRules

// Start real-time network monitoring

WHILE Network is Active

// Capture network traffic

Capture Incoming Packet

// Extract packet information

Extract:

SourceIP

DestinationIP

SourcePort

DestinationPort

Protocol

PacketContent

ThreatDetected = FALSE

// Check packet against known attack signatures

FOR each Signature in SignatureDatabase

IF PacketContent matches Signature THEN

ThreatDetected = TRUE

ThreatType = "Known Attack"

BREAK

ENDIF

END FOR

```
// Check against security rules
```

```
IF SourceIP is blocked THEN
```

```
    ThreatDetected = TRUE
```

```
    ThreatType = "Blocked Source"
```

```
ELSE IF DestinationPort is unauthorized THEN
```

```
    ThreatDetected = TRUE
```

```
    ThreatType = "Unauthorized Port"
```

```
ELSE IF PacketContent contains malicious pattern THEN
```

```
    ThreatDetected = TRUE
```

```
    ThreatType = "Malicious Request"
```

```
ENDIF
```

```
// Take action against malicious traffic
```

```
IF ThreatDetected = TRUE THEN
```

```
    DROP Packet
```

Log:

Time

SourceIP

DestinationIP

SourcePort

DestinationPort

Protocol

ThreatType

Generate Security Alert

Display "Malicious Traffic Blocked"

ELSE

ALLOW Packet

Display "Traffic Allowed"

ENDIF

END WHILE